



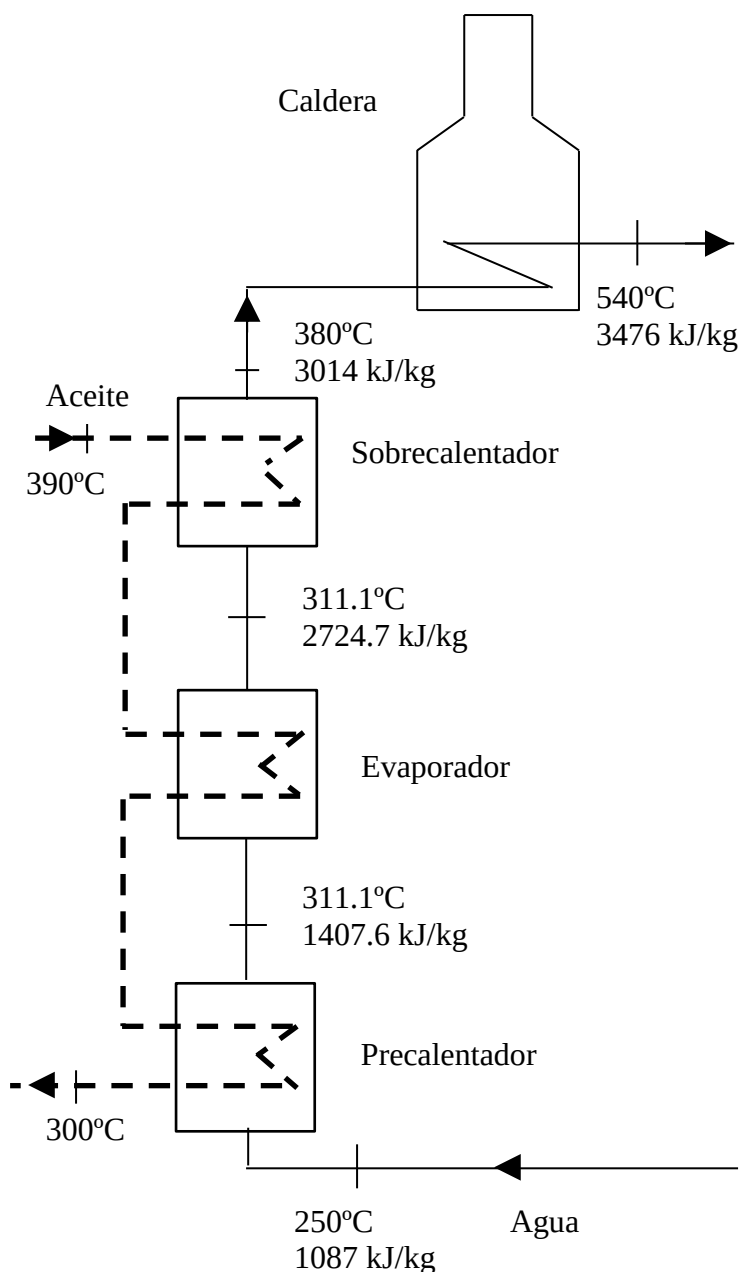
EPS. UC3M
Departamento de INGENIERÍA TÉRMICA Y DE FLUIDOS
ASIGNATURA: CALOR Y FRÍO INDUSTRIAL
14 Junio 2018

APELLIDOS:

NOMBRE:

Tiempo: 1 hora

El sistema de generación de vapor del ciclo de potencia de una central termosolar consiste en tres intercambiadores de calor de carcasa y tubos y una caldera de apoyo. Por los intercambiadores de calor circula como fluido caliente un aceite, que es un líquido orgánico de baja viscosidad y como fluido frío agua, que es el fluido de trabajo del ciclo de potencia. En la figura inferior se muestra un esquema con las temperaturas y entalpías de los fluidos en los diferentes puntos del proceso. El aceite tiene un calor específico de $2.4 \text{ kJ/kg}\cdot\text{K}$. El gasto de agua en los intercambiadores es de 15 kg/s . El combustible de la caldera de apoyo es un gas cuya composición puede aproximarse al metano (CH_4), con un poder calorífico inferior de 40 MJ/Nm^3 . El combustible se quema en la caldera con un exceso de aire del 10% y la temperatura de los humos a la salida es de 400°C y su calor específico $1.4 \text{ kJ/Nm}^3\cdot\text{K}$. Calcule el coste del evaporador y el caudal volumétrico de combustible que se quema en la caldera. El aire ambiente está a 20°C y se compone de 3.76 Nm^3 de N_2 por cada Nm^3 de O_2 .





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Tabla de costes intercambiadores de calor de carcasa y tubos:

$\dot{Q}/\Delta T$ (W/K)	Cold Side Fluid	Parameter	Hot Side Fluid								
			Low Pressure Gas (< 1 bar)	Medium Pressure Gas (20 bar)	High Pressure Gas (150 bar)	Process Water	Low Viscosity Organic Liquid	High Viscosity Liquid	Condensing Steam	Condensing Hydrocarbon	Condensing Hydrocarbon with Inert Gas
30 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	55 1.11	93 0.76	120 1.06	102 0.73	99 0.74	63 0.99	107 0.71	100 0.73	86 0.80
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	93 0.76	300 0.37	350 0.62	500 0.28	375 0.33	120 0.63	530 0.28	388 0.32	240 0.42
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	120 1.06	350 0.62	400 0.94	600 0.40	450 0.53	200 0.73	600 0.40	400 0.62	300 0.62
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	105 0.71	484 0.29	600 0.40	938 0.23	714 0.25	142 0.56	1607 0.19	764 0.24	345 0.34
	Low Viscosity Organic Liquid	U (W/m ² K) C (£/(W/K))	99 0.74	375 0.33	450 0.53	600 0.27	500 0.38	130 0.59	818 0.24	524 0.28	286 0.38
	High Viscosity Liquid	U (W/m ² K) C (£/(W/K))	68 0.94	138 0.57	200 0.73	161 0.52	153 0.53	82 0.83	173 0.50	155 0.53	124 0.60
	Boiling Water	U (W/m ² K) C (£/(W/K))	105 0.71	467 0.29	550 0.49	875 0.23	677 0.25	140 0.56	1432 0.20	722 0.25	336 0.35
	Boiling Organic Liquid	U (W/m ² K) C (£/(W/K))	99 0.74	375 0.33	450 0.53	600 0.27	500 0.28	130 0.59	818 0.24	524 0.28	286 0.38
100 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	55 0.95	93 0.58	120 0.93	102 0.54	99 0.55	63 0.83	107 0.52	100 0.55	86 0.62
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	93 0.58	300 0.23	350 0.35	429 0.18	375 0.20	120 0.47	530 0.16	388 0.19	240 0.27
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	120 0.93	350 0.35	400 0.58	600 0.24	450 0.28	200 0.64	600 0.24	400 0.32	300 0.39
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	105 0.52	484 0.17	600 0.24	938 0.116	714 0.134	142 0.41	1607 0.086	764 0.129	345 0.21
	Low Viscosity Organic Liquid	U (W/m ² K) C (£/(W/K))	99 0.55	375 0.20	450 0.28	609 0.145	500 0.162	130 0.44	818 0.125	524 0.158	286 0.24
	High Viscosity Liquid	U (W/m ² K) C (£/(W/K))	68 0.77	138 0.42	200 0.64	161 0.37	153 0.38	82 0.65	173 0.35	155 0.38	124 0.45
	Boiling Water	U (W/m ² K) C (£/(W/K))	105 0.52	467 0.168	550 0.26	875 0.121	677 0.137	140 0.41	1432 0.091	722 0.133	336 0.22
	Boiling Organic Liquid	U (W/m ² K) C (£/(W/K))	99 0.55	375 0.20	450 0.28	600 0.146	500 0.162	130 0.44	818 0.125	524 0.158	286 0.24
1 000 000	Low Pressure Gas (< 1 bar)	U (W/m ² K) C (£/(W/K))	55 0.93	93 0.55	120 0.93	102 0.50	99 0.52	63 0.81	107 0.48	100 0.51	86 0.60
	Medium Pressure Gas (20 bar)	U (W/m ² K) C (£/(W/K))	93 0.55	300 0.171	350 0.32	500 0.103	375 0.137	120 0.43	530 0.098	388 0.132	240 0.21
	High Pressure Gas (150 bar)	U (W/m ² K) C (£/(W/K))	120 0.93	350 0.32	400 0.58	600 0.19	450 0.25	200 0.56	600 0.19	400 0.28	300 0.37
	Treated Cooling Water	U (W/m ² K) C (£/(W/K))	105 0.49	484 0.106	600 0.19	938 0.058	714 0.074	142 0.36	1607 0.037	764 0.069	345 0.148
	Low Viscosity Organic Liquid	U (W/m ² K) C (£/(W/K))	99 0.52	375 0.137	450 0.25	600 0.087	500 0.103	130 0.39	818 0.065	524 0.099	286 0.179
	High Viscosity Liquid	U (W/m ² K) C (£/(W/K))	68 0.75	138 0.37	200 0.56	161 0.32	153 0.34	82 0.62	173 0.30	155 0.33	124 0.41
	Boiling Water	U (W/m ² K) C (£/(W/K))	105 0.49	467 0.11	550 0.20	875 0.061	677 0.077	140 0.37	1432 0.04	722 0.073	336 0.152
	Boiling Organic Liquid	U (W/m ² K) C (£/(W/K))	99 0.52	375 0.137	450 0.25	600 0.087	500 0.103	130 0.39	818 0.065	524 0.098	280 0.179

Solución: 71610 libras, 0.204 m³/s

Datos:

$$\text{kJ} := 1000\text{J}$$

$$\text{MJ} := 10^6\text{J}$$

$$m_v := 15 \frac{\text{kg}}{\text{s}}$$

$$T_{\text{aeprec}} := 250^\circ\text{C}$$

$$T_{\text{acesobrec}} := 390^\circ\text{C}$$

$$T_{\text{vssobrec}} := 380^\circ\text{C}$$

$$T_{\text{sat}} := 311.1^\circ\text{C}$$

$$T_{\text{acsprec}} := 300^\circ\text{C}$$

$$h_{\text{lv}} := 1317.1 \frac{\text{kJ}}{\text{kg}}$$

$$c_{\text{pac}} := 2.4 \frac{\text{kJ}}{\text{kg}\cdot\text{K}}$$

$$h_{\text{eprec}} := 1087 \frac{\text{kJ}}{\text{kg}}$$

$$h_{\text{lsat}} := 1407.6 \frac{\text{kJ}}{\text{kg}}$$

$$h_{\text{vsat}} := 2724.7 \frac{\text{kJ}}{\text{kg}}$$

$$n := 1.1$$

$$h_{\text{vssobrec}} := 3014 \frac{\text{kJ}}{\text{kg}}$$

$$\text{PCI} := 40 \frac{\text{MJ}}{\text{m}^3}$$

$$T_{\text{hs}} := 400^\circ\text{C}$$

$$h_{\text{vsald}} := 3476 \frac{\text{kJ}}{\text{kg}}$$

$$c_{\text{ph}} := 1.35 \frac{\text{kJ}}{\text{m}^3\text{K}}$$

$$T_{\text{amb}} := 20^\circ\text{C}$$

$$T_{\text{vsalcald}} := 540^\circ\text{C}$$

Potencias intercambiadores de calor:

$$Q_{\text{prec}} := m_v \cdot (h_{\text{lsat}} - h_{\text{eprec}}) = 4.809 \times 10^6 \text{W}$$

$$Q_{\text{evap}} := m_v \cdot h_{\text{lv}} = 1.976 \times 10^7 \text{W}$$

$$Q_{\text{sobrec}} := m_v \cdot (h_{\text{vssobrec}} - h_{\text{vsat}}) = 4.34 \times 10^6 \text{W}$$

$$Q_{\text{IC}} := Q_{\text{prec}} + Q_{\text{evap}} + Q_{\text{sobrec}} = 2.89 \times 10^7 \text{W}$$

$$m_{\text{ac}} := \frac{Q_{\text{IC}}}{c_{\text{pac}} \cdot (T_{\text{acesobrec}} - T_{\text{acsprec}})} = 133.819 \frac{\text{kg}}{\text{s}}$$

Temperaturas del aceite:

$$T_{\text{acssobrec}} := T_{\text{acesobrec}} - \frac{Q_{\text{sobrec}}}{m_{\text{ac}} \cdot c_{\text{pac}}} = 649.638 \cdot \text{K}$$

$$\frac{T_{\text{acssobrec}}}{\text{K}} - 273.1 = 376.538$$

$$T_{aceprec} := T_{acsprec} + \frac{Q_{prec}}{m_{ac} \cdot c_{pac}} = 588.124 \text{ K} \quad \frac{T_{aceprec}}{K} - 271.15 = 316.974$$

Diferencia de temperatura media en los intercambiadores de calor:

El precalentador y el sobrecalentador son IC en contracorriente: F=1

Para el evaporador: F=1

$$\Delta T_{lmsobrec} := \frac{(T_{acesobrec} - T_{vssobrec}) - (T_{acssobrec} - T_{sat})}{\ln \left[\frac{(T_{acesobrec} - T_{vssobrec})}{(T_{acssobrec} - T_{sat})} \right]} = 29.497 \text{ K}$$

$$\Delta T_{lmevap} := \frac{(T_{acssobrec} - T_{sat}) - (T_{aceprec} - T_{sat})}{\ln \left[\frac{(T_{acssobrec} - T_{sat})}{(T_{aceprec} - T_{sat})} \right]} = 21.766 \text{ K}$$

$$\Delta T_{lmprec} := \frac{(T_{aceprec} - T_{sat}) - (T_{acsprec} - T_{aeprec})}{\ln \left[\frac{(T_{aceprec} - T_{sat})}{(T_{acsprec} - T_{aeprec})} \right]} = 18.033 \text{ K}$$

$$Q \Delta T_{sobrec} := \frac{Q_{sobrec}}{\Delta T_{lmsobrec}} = 1.471 \times 10^5 \cdot \frac{W}{K}$$

$$Q \Delta T_{evap} := \frac{Q_{evap}}{\Delta T_{lmevap}} = 9.077 \times 10^5 \cdot \frac{W}{K}$$

$$Q \Delta T_{prec} := \frac{Q_{prec}}{\Delta T_{lmprec}} = 2.667 \times 10^5 \cdot \frac{W}{K}$$

$$Q \Delta T_1 := 100000 \frac{W}{K} \quad C_1 := 0.137$$

$$Q \Delta T_2 := 1000000 \frac{W}{K} \quad C_2 := 0.077$$

$$C_{sobrecunit} := e^{\left(\ln(C_1) + \ln \left(\frac{C_1}{C_2} \right) \cdot \frac{\ln \left(\frac{Q \Delta T_{sobrec}}{Q \Delta T_1} \right)}{\ln \left(\frac{Q \Delta T_1}{Q \Delta T_2} \right)} \right)} = 0.124$$

$$C_{\text{sobrec}} := C_{\text{sobrecunit}} \cdot \frac{Q\Delta T_{\text{sobrec}}}{\frac{W}{K}} = 1.83 \times 10^4$$

$$C_{\text{evapunit}} := e^{\left(\ln(C_1) + \ln\left(\frac{C_1}{C_2}\right) \cdot \frac{\ln\left(\frac{Q\Delta T_{\text{evap}}}{Q\Delta T_1}\right)}{\ln\left(\frac{Q\Delta T_1}{Q\Delta T_2}\right)} \right)} = 0.079$$

$$C_{\text{evap}} := C_{\text{evapunit}} \cdot \frac{Q\Delta T_{\text{evap}}}{\frac{W}{K}} = 7.161 \times 10^4$$

$$C_{\text{precunit}} := e^{\left(\ln(C_1) + \ln\left(\frac{C_1}{C_2}\right) \cdot \frac{\ln\left(\frac{Q\Delta T_{\text{prec}}}{Q\Delta T_1}\right)}{\ln\left(\frac{Q\Delta T_1}{Q\Delta T_2}\right)} \right)} = 0.107$$

$$C_{\text{prec}} := C_{\text{precunit}} \cdot \frac{Q\Delta T_{\text{prec}}}{\frac{W}{K}} = 2.858 \times 10^4$$

Caldera de apoyo:

$$V_{\text{BC}} := 1 + 2 + 0.1 \cdot 2 + 1.1 \cdot 2 \cdot 3.76 = 11.472$$

$$\eta := 1 - \frac{V_{\text{BC}} \cdot c_{\text{ph}} \cdot (T_{\text{hs}} - T_{\text{amb}})}{\text{PCI}} = 0.853$$

$$Q_{\text{cald}} := m_v \cdot (h_{\text{vsald}} - h_{\text{vssobrec}}) = 6.93 \times 10^6 \text{ W}$$

$$V_c := \frac{Q_{\text{cald}}}{\eta \cdot \text{PCI}} = 0.203 \frac{\text{m}^3}{\text{s}}$$